

ECON1050 Tutorial 12

Matrix Algebra III

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Today's Plan

- ▶ Recap: key ideas from Lecture 12
- ▶ Tutorial questions: Q1–Q5

Recap: Cramer's Rule

Core idea

For a system

$$\mathbf{Ax} = \mathbf{b},$$

Cramer's Rule uses determinants to solve for one variable at a time.

Formula

If $|\mathbf{A}| \neq 0$, then

$$x_k = \frac{|\mathbf{A}_k|}{|\mathbf{A}|},$$

where $\mathbf{A}_k$ is formed by replacing the k th column of $\mathbf{A}$ with the constants vector $\mathbf{b}$.

When useful

Cramer's Rule is useful when we only need one endogenous variable.

Recap: Cramer's Rule for a 3×3 system

System

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Replacement matrices

$$x_1 = \frac{|\mathbf{A}_1|}{|\mathbf{A}|}, \quad x_2 = \frac{|\mathbf{A}_2|}{|\mathbf{A}|}, \quad x_3 = \frac{|\mathbf{A}_3|}{|\mathbf{A}|}.$$

Each $\mathbf{A}_k$ replaces column k of $\mathbf{A}$ with $\mathbf{b}$.

Recap: Minors and cofactors

Minor

For a square matrix $\mathbf{A}$, the minor M_{ij} is the determinant of the submatrix obtained by deleting row i and column j .

Cofactor

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

A cofactor exists for every element of a square matrix.

Matrix of cofactors

The matrix of cofactors is denoted by $\mathbf{C}^+$.

Recap: The adjoint method

Adjoint / adjugate

The transpose of the matrix of cofactors is called the adjoint:

$$\text{adj}(\mathbf{A}) = (\mathbf{C}^+)'.$$

Inverse using the adjoint

If $|\mathbf{A}| \neq 0$, then

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj}(\mathbf{A}).$$

Procedure

Find $|\mathbf{A}|$, find all cofactors, transpose the cofactor matrix, then divide by $|\mathbf{A}|$.

Recap: Solving systems using the inverse

Inverse method

If

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

and $\mathbf{A}^{-1}$ exists, then

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

Non-singularity condition

The inverse exists if and only if

$$|\mathbf{A}| \neq 0.$$

Tutorial Q1

Question 1

Solve the system using Cramer's Rule:

$$\begin{bmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 8 \end{bmatrix}.$$

Tutorial Q1: Cramer's Rule

Knowledge used

- Use Cramer's Rule:

$$x_k = \frac{|\mathbf{A}_k|}{|\mathbf{A}|}.$$

- $\mathbf{A}_k$ replaces the k th column of $\mathbf{A}$ with $\mathbf{b}$.

Solution

$$|\mathbf{A}| = \begin{vmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{vmatrix} = -49.$$

$$|\mathbf{A}_1| = 49, \quad |\mathbf{A}_2| = -49, \quad |\mathbf{A}_3| = -98.$$

Therefore,

$$x_1 = \frac{49}{-49} = -1, \quad x_2 = \frac{-49}{-49} = 1, \quad x_3 = \frac{-98}{-49} = 2.$$

$$(x_1, x_2, x_3) = (-1, 1, 2).$$

Tutorial Q2

Question 2

Consider the model

$$Y = C + \bar{I} + \bar{G}, \quad C = \alpha + \beta(Y - T), \quad T = \gamma + \delta Y.$$

Write the model in matrix form and solve for T^* using Cramer's Rule.

Tutorial Q2: Matrix form

Knowledge used

- ▶ Rearrange equations so that endogenous variables are on the left.
- ▶ Write the system as $\mathbf{Ax} = \mathbf{b}$.

Solution

Rearrange:

$$Y - C = \bar{I} + \bar{G}, \quad -\beta Y + C + \beta T = \alpha, \quad -\delta Y + T = \gamma.$$

Hence

$$\begin{bmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{bmatrix} \begin{bmatrix} Y \\ C \\ T \end{bmatrix} = \begin{bmatrix} \bar{I} + \bar{G} \\ \alpha \\ \gamma \end{bmatrix}.$$

Tutorial Q2: Solving for tax revenue

Knowledge used

- Use Cramer's Rule for the variable T .
- Replace the T column of $\mathbf{A}$ with the constants vector.

Solution

$$|\mathbf{A}| = \begin{vmatrix} 1 & -1 & 0 \\ -\beta & 1 & \beta \\ -\delta & 0 & 1 \end{vmatrix} = 1 - \beta(1 - \delta).$$

$$|\mathbf{A}_T| = \begin{vmatrix} 1 & -1 & \bar{I} + \bar{G} \\ -\beta & 1 & \alpha \\ -\delta & 0 & \gamma \end{vmatrix} = \delta(\alpha + \bar{I} + \bar{G}) + \gamma(1 - \beta).$$

Thus

$$T^* = \frac{\delta(\alpha + \bar{I} + \bar{G}) + \gamma(1 - \beta)}{1 - \beta(1 - \delta)}.$$

Question 3

Find the inverse using the adjoint method:

$$(1) \begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix}, \quad (2) \begin{bmatrix} 11 & -6 \\ 18 & -10 \end{bmatrix},$$

$$(3) \begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix}.$$

Tutorial Q3: Adjoint method I

Knowledge used

- Use

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj}(\mathbf{A}).$$

- For 2×2 matrices, compute determinant and cofactors directly.

Solution

For

$$\mathbf{A} = \begin{bmatrix} 5 & 2 \\ 1 & -1 \end{bmatrix}, \quad |\mathbf{A}| = -7.$$

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} -1 & -2 \\ -1 & 5 \end{bmatrix}.$$

Hence

$$\mathbf{A}^{-1} = \begin{bmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{1}{7} & -\frac{5}{7} \end{bmatrix}.$$

Tutorial Q3: Adjoint method II

Knowledge used

- ▶ The same adjoint formula applies to any non-singular square matrix.
- ▶ Check $|\mathbf{A}| \neq 0$ before finding the inverse.

Solution

For

$$\mathbf{B} = \begin{bmatrix} 11 & -6 \\ 18 & -10 \end{bmatrix}, \quad |\mathbf{B}| = -2.$$

$$\text{adj}(\mathbf{B}) = \begin{bmatrix} -10 & 6 \\ -18 & 11 \end{bmatrix}.$$

Therefore,

$$\mathbf{B}^{-1} = \begin{bmatrix} 5 & -3 \\ 9 & -\frac{11}{2} \end{bmatrix}.$$

Tutorial Q3: Adjoint method III

Knowledge used

- For a 3×3 matrix, compute all cofactors.
- Then transpose the cofactor matrix to get the adjoint.

Solution

For

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -2 \\ 4 & -2 & 1 \\ 1 & 2 & -10 \end{bmatrix}, \quad |\mathbf{A}| = -2.$$

The adjoint is

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} 18 & -4 & -4 \\ 41 & -8 & -9 \\ 10 & -2 & -2 \end{bmatrix}.$$

Thus

$$\mathbf{A}^{-1} = \begin{bmatrix} -9 & 2 & 2 \\ -\frac{41}{2} & 4 & \frac{9}{2} \\ -5 & 1 & 1 \end{bmatrix}.$$

Question 4

Solve the system using the inverse:

$$2x_1 + 3x_2 - x_3 = -1,$$

$$5x_1 + x_2 + x_3 = -2,$$

$$-x_1 - x_2 + 4x_3 = 8.$$

Tutorial Q4: Inverse method

Knowledge used

- Write the system as $\mathbf{Ax} = \mathbf{b}$.
- If $\mathbf{A}^{-1}$ exists, then $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

Solution

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & -1 \\ 5 & 1 & 1 \\ -1 & -1 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ -2 \\ 8 \end{bmatrix}.$$

$$\mathbf{A}^{-1} = \begin{bmatrix} -\frac{5}{49} & \frac{11}{49} & -\frac{4}{49} \\ \frac{3}{7} & -\frac{1}{7} & \frac{1}{7} \\ \frac{4}{49} & \frac{1}{49} & \frac{13}{49} \end{bmatrix}.$$

Hence

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}.$$

Therefore,

$$(x_1, x_2, x_3) = (-1, 1, 2).$$

Question 5

Solve the system:

$$2x + y + z = 0, \quad 4x + 3y + 2z = 2, \quad 2x - y - 3z = 0.$$

Use both the inverse method and Cramer's Rule.

Tutorial Q5: Inverse method

Knowledge used

- ▶ Use $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.
- ▶ The inverse can be found using the adjoint method.

Solution

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 2 \\ 2 & -1 & -3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

$$\mathbf{A}^{-1} = \begin{bmatrix} \frac{7}{8} & -\frac{1}{4} & \frac{1}{8} \\ -2 & 1 & 0 \\ \frac{5}{4} & -\frac{1}{2} & -\frac{1}{4} \end{bmatrix}.$$

Thus

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \begin{bmatrix} -\frac{1}{2} \\ 2 \\ -1 \end{bmatrix}.$$

$$(x, y, z) = \left(-\frac{1}{2}, 2, -1\right).$$

Tutorial Q5: Cramer's Rule

Knowledge used

- ▶ Cramer's Rule gives the same solution as the inverse method.
- ▶ Use $x_k = |\mathbf{A}_k|/|\mathbf{A}|$.

Solution

$$|\mathbf{A}| = -8, \quad |\mathbf{A}_1| = 4, \quad |\mathbf{A}_2| = -16, \quad |\mathbf{A}_3| = 8.$$

Therefore,

$$x = \frac{4}{-8} = -\frac{1}{2}, \quad y = \frac{-16}{-8} = 2, \quad z = \frac{8}{-8} = -1.$$

So

$$(x, y, z) = \left(-\frac{1}{2}, 2, -1\right).$$

Key takeaways

- ▶ Cramer's Rule solves variables using ratios of determinants.
- ▶ The adjoint method gives $\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj}(\mathbf{A})$ when $|\mathbf{A}| \neq 0$.
- ▶ Inverse matrices solve systems through $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

A Few Final Words

- ▶ Thank you all for your company and support throughout this semester.
- ▶ Next semester, I will most likely be tutoring ECON2050 Mathematical Economics. Feel free to enrol if you are interested.
- ▶ Wish you all the best in your future studies.
- ▶ Hope you continue to keep your passion for learning and your curiosity alive.